

Jacob Berger

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EDUCATION

Oregon State University Honors College
Bachelor of Science in Applied Computer Science
Bachelor of Science in Computational Mathematics

Corvallis, OR
September 2022 – June 2026

- **Concentration:** Artificial Intelligence | **GPA:** 3.9+ | Accelerated Master's Program
- **Relevant Coursework:** Analysis of Algorithms, Machine Learning, Linear Algebra II. **Graduate:** Deep Learning, Operating Systems II, Numerical ODEs, Math of Data Science, Computational Number Theory

RESEARCH

Honors Thesis: Dynamic Constraints for LLMs
Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

Corvallis, OR
June 2025 – Present

- Leading an honors thesis project to develop a dynamic constraint system for large language models, improving text generation for applications in computational semantics.

Senior Capstone Project: AI for Home Automation
Project Team Member

Corvallis, OR
September 2025 – Present

- Collaborating in a senior design team to integrate Large Language Models with the Home Assistant IoT platform, enabling users to control devices and create complex automations using natural language.

EXPERIENCE

NLP SaaS Startup
Language Model and Software Engineer

Portland, OR
March 2023 – Present

- Architected end-to-end data pipelines for training and fine-tuning specialized large language models.
- Led and co-authored the team's approved Phase I Small Business Innovation Research (SBIR) proposal.
- Engineered and deployed cloud infrastructure on AWS, GCP, and Vercel to train and host services.

MU Lanes and Games
Head Mechanic

Corvallis, OR
December 2022 – Present

- Managed maintenance projects and diagnostics for complex electromechanical systems.

MU Student Activities and Advisory Board
Vice-Chairman

Corvallis, OR
June 2024 – Present

- Oversaw budget preparation and facilitated strategic board meetings for OSU's Memorial Union.

SKILLS

- **Languages:** Python, C++, Java, Javascript, Matlab, Bash/Shell, x86 Assembly
- **Machine Learning:** PyTorch, HuggingFace Transformers, LLM Fine-tuning (LoRA), RAG Pipelines
- **Systems & Infrastructure:** Linux (Debian), Proxmox VE, Virtualization (LXC/KVM), GPU Passthrough
- **Cloud & DevOps:** AWS (EC2/S3), Google Cloud Platform (GCP), Git, Vercel
- **Languages (Human):** Fluent in Spanish (9 years of language immersion education)

AWARDS

- **Putnam Math Competition Award:** Top 3 Finisher at Oregon State University.
- **Finley Academic Excellence Scholarship:** \$20,000 scholarship recipient for academic merit at OSU.

EXTRACURRICULARS

- **Competitive Math:** Competitor in Oregon State Putnam and COMAP Math Modeling competitions.
- **Competitive Sports:** Founded OSU's national collegiate bowling team (All-Star qualifier); served as Rose City Rowing Club team leader (2-time national qualifier).